TOPIC 09

Confidence Intervals

From a point estimate to a range of plausible values: z, t and proportion intervals, interpretation, sample size and difference of means

WHAT YOU WILL LEARN

- Explain what a confidence interval and a confidence level mean, and interpret them correctly
- Build intervals for a mean (σ known with z, σ unknown with t) and for a proportion
- Read z and t critical values from tables and software
- See how confidence level, sample size and variability change the width
- Plan the sample size for a target margin of error
- Construct an interval for a difference of two means

1. The problem with a single number

In Topic 08 we saw that a **point estimate** is a single best guess with no sign of how wrong it might be. Suppose you visit 5% of the restaurants in London and find the average meal costs **£22.50**. You may be close, but the true average is unlikely to be exactly 22.50; somewhere between £20 and £25 is more believable. The interval **[20, 25]** captures the reality with far more honesty.

DEFINITION **Confidence interval (CI)**

A **confidence interval** is a range of values, computed from sample data, that is likely to contain the true population parameter. The **margin of error** is the "±" part: how far the interval reaches on each side of the estimate. The interval has the form

$$\text{point estimate} \pm \text{margin of error} = \text{point estimate} \pm (\text{reliability factor}) \times \text{SE}$$

In the notation from the lecture slides the interval for a mean is [$\bar{x}$ − (reliability factor) × σ/√n , $\bar{x}$ + (reliability factor) × σ/√n]. The estimate sits in the middle; the reliability factor depends on how confident we want to be; the standard error (Topic 07) measures the precision of the estimate.

2. Confidence level and α

The **confidence level** is written **(1 − α)**, where **α** (alpha) is a number between 0 and 1 called the **significance level**. A 95% confidence level means α = 0.05; 99% means α = 0.01.

Confidence level (1 − α)	α	α/2 (each tail)	Critical value z*
90%	0.10	0.05	1.645
95%	0.05	0.025	1.960
99%	0.01	0.005	2.576

COMMON MISTAKE **Careful with the words**

The 5% is **α** (the significance level, the chance of 'missing'), not the confidence level. The confidence level is 95% = 1 − α. A 95% CI leaves 5% in total, split into 2.5% in each tail.

You can never be 100% sure unless you survey the entire population. A higher confidence level is bought at the price of a wider interval (Section 8).

3. How to interpret a confidence interval

Here is where almost everyone slips. The parameter μ is **fixed**. It is the *interval* that changes from sample to sample. The 95% describes the **method**, not one particular interval:

KEY IDEA **Correct meaning**

If we repeated the sampling many times and built an interval each time, about **95% of those intervals** would contain the true parameter. For the one interval we actually computed, we have '95% confidence' that it is one of the good ones.

COMMON MISTAKE **Wrong interpretations**

- 'There is a 95% probability that μ is in [94,832, 105,568].' μ is fixed; the interval either contains it or does not. The probability statement is about the procedure.
- '95% of the data lies in this interval.' No: the interval is about the **mean**, not individual observations.
- '95% of future sample means will fall inside this interval.' Not guaranteed: that depends on where this particular sample mean happened to land.

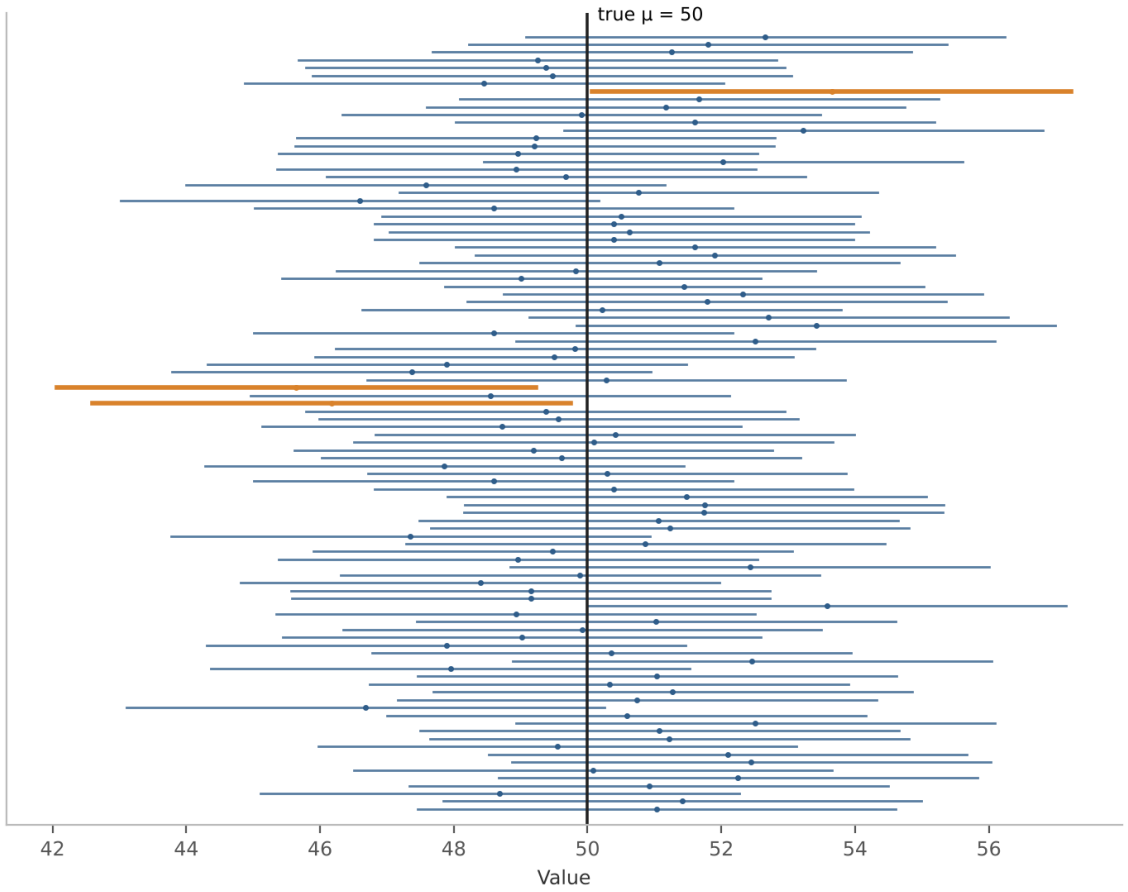


Figure 1. 100 different samples (n = 30) from a population with μ = 50, σ = 10, each giving a 95% interval. Blue intervals contain μ (97 of 100); orange ones (3) miss it. In the long run 5 in 100 miss; any single batch of 100 varies around that.

4. Case 1: mean, population standard deviation known (z interval)

When the population is normal (or n is large, by the CLT) and σ is known, $\bar{x}$ has a normal sampling distribution with standard error σ/√n. Then

$$\left[\, \bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \;,\quad \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \,\right]$$

$\mathbf{z_{\alpha/2}}$ is the **critical value**: the z-score that leaves α/2 in the upper tail. We find it from the standard normal table (Topic 04).

Worked example (the lecture's data scientist salaries)

You want to know what salary a data scientist earns. The population standard deviation of salaries is σ = $15,000 and salaries are normally distributed. A sample of n = 30 salaries has mean $\bar{x}$ = **$100,200**. The standard error is SE = 15,000/√30 = **$2,739**.

WORKED EXAMPLE **95% confidence interval**

- α = 0.05, so α/2 = 0.025. We need the z with area 1 − 0.025 = **0.975** to its left.
- In the z-table, find 0.9750: row 1.9, column 0.06. So z = 1.9 + 0.06 = **1.96**.
- Margin of error = 1.96 × 2,738.6 = $5,368.
- CI = 100,200 ± 5,368 (SE kept unrounded at 2,738.6) = **[$94,832 , $105,568]**.

WORKED EXAMPLE **99% confidence interval**

- α = 0.01, α/2 = 0.005, area to the left = **0.995**.
- The table gives z = 2.5 + 0.08 = **2.58** (exactly 2.576).
- Margin of error = 2.576 × 2,738.6 = $7,054.
- CI = **[93,146 , 107,254]**.

Confidence	z*	Margin of error	Interval	Width
95%	1.960	5,368	94,832 to 105,568	10,735
99%	2.576	7,054	93,146 to 107,254	14,108

This is the **trade-off** between confidence and precision: the 99% interval is wider than the 95% interval. We are more confident because the net is wider, but less informative.

NOTE **Two typos to avoid in the class material**

- The class notes wrote '1 − 0.025 = 0.0975'. The correct subtraction is 1 − 0.025 = **0.975**. Looking up 0.0975 in the table would give the wrong z.
- The class slides (slides 19 and 20) print the 99% interval as [93,135, 107,206]. That is not symmetric about 100,200. With their z = 2.58 the upper limit should be about **107,266**; with z = 2.576, as above, the interval is [93,146, 107,254].

5. Case 2: mean, σ unknown (t interval)

In practice we almost never know σ. We replace it by the sample standard deviation s, but this adds extra uncertainty. For small samples the standardized statistic no longer follows the normal curve; it follows **Student's t distribution** (published by W. S. Gosset under the pen-name 'Student', 1908).

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{n-1} \qquad \Longrightarrow \qquad \left[\bar{x} - t_{\alpha/2,\, n-1}\frac{s}{\sqrt{n}} \;,\;\; \bar{x} + t_{\alpha/2,\, n-1}\frac{s}{\sqrt{n}} \right]$$

- The t distribution is symmetric and bell-shaped like the normal but has **heavier tails**, which is the price of estimating σ.
- Its shape depends on the **degrees of freedom** df = n − 1. As df grows, t approaches the standard normal.
- t critical values are always **larger** than z critical values, so t intervals are wider, especially for small n.

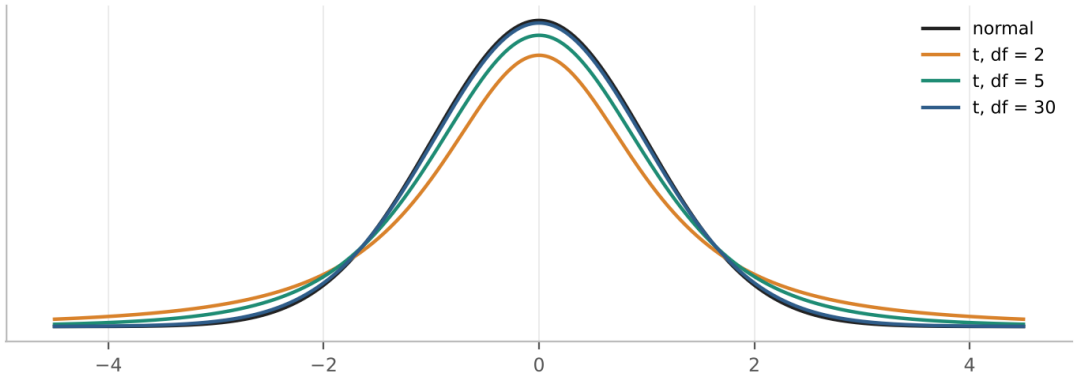


Figure 2. Student's t versus the normal. Lower df means fatter tails; by df = 30 the curves almost coincide.

Table of t critical values (two-sided). Find your df in the left column of a block, then read across for 90%, 95% or 99% confidence. The last row (df = ∞) is the normal z* from Topic 04. If your df is not listed (for example 68), use the next smaller listed df (60), which is slightly conservative, or ask software.

df	90%	95%	99%	df	90%	95%	99%	df	90%	95%	99%
1	6.314	12.706	63.657	13	1.771	2.160	3.012	24	1.711	2.064	2.797
2	2.920	4.303	9.925	14	1.761	2.145	2.977	25	1.708	2.060	2.787
3	2.353	3.182	5.841	15	1.753	2.131	2.947	26	1.706	2.056	2.779
4	2.132	2.776	4.604	16	1.746	2.120	2.921	27	1.703	2.052	2.771
5	2.015	2.571	4.032	17	1.740	2.110	2.898	28	1.701	2.048	2.763
6	1.943	2.447	3.707	18	1.734	2.101	2.878	29	1.699	2.045	2.756
7	1.895	2.365	3.499	19	1.729	2.093	2.861	30	1.697	2.042	2.750
8	1.860	2.306	3.355	20	1.725	2.086	2.845	40	1.684	2.021	2.704
9	1.833	2.262	3.250	21	1.721	2.080	2.831	60	1.671	2.000	2.660
10	1.812	2.228	3.169	22	1.717	2.074	2.819	120	1.658	1.980	2.617
11	1.796	2.201	3.106	23	1.714	2.069	2.807	∞	1.645	1.960	2.576
12	1.782	2.179	3.055								

WORKED EXAMPLE **Exam scores, σ unknown**

A sample of n = 16 students has $\bar{x} = 72$ and s = 10. SE = 10/√16 = 2.50. With df = 15, t* = **2.131** (a z interval would wrongly use 1.960). Margin of error = 2.131 × 2.50 = 5.33. The 95% CI is **[66.67 , 77.33]**.

KEY IDEA **Which to use for a mean?**

- **σ known** (rare): z interval.
- **σ unknown** (usual): t interval with n − 1 df, for any n. For large n the two agree closely, which is why many books use z when n ≥ 30. Software always uses t.
- Data strongly non-normal and n small: neither is trustworthy; consider the bootstrap or a transformation.

6. Case 3: a proportion

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Valid when $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$. We use $\hat{p}$ in the standard error because the true p is unknown.

WORKED EXAMPLE **A poll**

In a survey of n = 1,000 voters, 520 support a measure, so $\hat{p} = 0.52$. SE = √(0.52 × 0.48/1000) = 0.0158. For 95%: margin of error = 1.96 × 0.0158 = 0.0310. CI = **[0.4890 , 0.5510]**, that is 48.9% to 55.1%. Because 50% lies inside, the poll cannot say the measure has majority support.

7. Planning the sample size

Because margin of error E = z* × σ/√n, we can solve for n to guarantee a chosen precision **before** collecting data:

$$n = \left(\frac{z_{\alpha/2}\,\sigma}{E}\right)^2 \qquad\qquad n = \frac{z_{\alpha/2}^2\,p(1-p)}{E^2}$$

The left formula is for a mean, the right for a proportion. Always **round up** to the next whole number.

- Salary mean to within $2,000 with 95% confidence, σ = 15,000: n = (1.96 × 15,000/2,000)² = 216.1, so **n = 217**.
- Poll proportion within ±3 points, 95% confidence. With no prior knowledge use the worst case p = 0.5: n = 1.96² × 0.25/0.03² = 1067.1, so **n = 1068**.

Note the square: halving the margin of error requires four times the sample size.

8. What controls the width?

If you...	The interval...	Because...
Raise confidence (95% → 99%)	gets **wider**	z* or t* increases
Increase n	gets **narrower**	SE = σ/√n decreases
Have more variable data (larger σ or s)	gets **wider**	SE increases

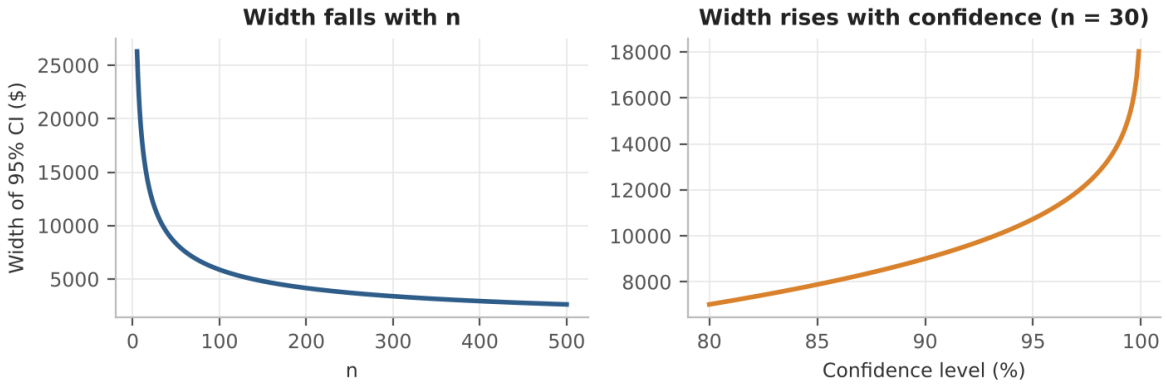


Figure 3. Width of the salary interval (σ = 15,000). More data tightens it; demanding more confidence widens it.

9. Difference between two means

To compare two groups we build a CI for **μ₁ − μ₂**. For independent samples with unknown variances (Welch's method):

$$(\bar{x}_1 - \bar{x}_2) \;\pm\; t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

The **Welch-Satterthwaite degrees of freedom** for t* are

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

Software computes this for you. It always lies between $\min(n_1, n_2) - 1$ (a safe, conservative lower bound you can use with a table) and $n_1 + n_2 - 2$ (the upper bound, which would be right if the variances were equal).

> **WORKED EXAMPLE** **Two teaching methods**
> Method A: $n = 40$, $\bar{x} = 82.4$, $s = 9.1$. Method B: $n = 35$, $\bar{x} = 78.1$, $s = 10.3$. Difference = 4.3. SE = $\sqrt{(9.1^2/40 + 10.3^2/35)}$ = **2.259**. Welch df ≈ 68.5, $t^* = 1.995$. CI = $4.3 \pm 1.995 \times 2.259$ = **[-0.21 , 8.81]**.

The interval **just includes 0** (its lower end is -0.21). So the data are consistent with no difference at all, even though Method A averaged 4.3 points higher. The evidence for Method A is suggestive but not conclusive; a hypothesis test (Topic 10) gives $t = 1.90$ and $p \approx 0.061$, the same message. Had the interval been entirely above 0, we would conclude Method A is better.

For **paired** data (before/after on the same people) compute each difference and use the one-sample t interval on those differences.

> **WORKED EXAMPLE** **Paired interval: blood pressure (same data as Topic 11)**
> Eight patients, systolic pressure before minus after: differences = [8, 5, 2, 7, 9, 5, 4, 4]. $\bar{d} = 5.50$, $s_d = 2.33$, SE = $2.33/\sqrt{8} = 0.82$. With df = 7, $t^* = 2.365$. Margin of error = $2.365 \times 0.82 = 1.95$. The 95% CI for the mean drop is **[3.55 , 7.45]**. It excludes 0, so the drop is real at the 5% level.

10. Which formula do I use?

Parameter	Conditions	Interval
Mean μ	σ known, normal data or large n	$\bar{x} \pm z^* \sigma/\sqrt{n}$
Mean μ	σ unknown (usual case)	$\bar{x} \pm t^* s/\sqrt{n}$, df = $n - 1$
Proportion p	$n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$	$\hat{p} \pm z^* \sqrt{(\hat{p}(1 - \hat{p})/n)}$
$\mu_1 - \mu_2$ (independent)	Two independent samples	$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$
Mean difference (paired)	Same units measured twice	$\bar{d} \pm t^* s_d/\sqrt{n}$

11. Common mistakes

- Interpreting 95% as 'there is a 95% probability μ is in this particular interval'.
- Using z when σ is unknown and n is small; use t.
- Confusing standard deviation with standard error in the formula.
- Reading z from the wrong tail, or using α instead of $\alpha/2$.
- Assuming a CI fixes a biased sample. It only reflects random error.
- Forgetting to round the required sample size **up**.

12. In Python

```
import numpy as np
from scipy import stats

# z interval, sigma known (data scientist salaries)
xbar, sigma, n = 100_200, 15_000, 30
print(stats.norm.interval(0.95, loc=xbar, scale=sigma/np.sqrt(n)))   # (94832, 105568)

# t interval, sigma unknown
data = np.array([68, 75, 80, 71, 66, 79, 74, 70, 77, 73])
print(stats.t.interval(0.95, df=len(data)-1, loc=data.mean(), scale=stats.sem(data)))

# proportion
phat, n = 0.52, 1000
print(stats.norm.interval(0.95, loc=phat, scale=np.sqrt(phat*(1-phat)/n)))
```

SUMMARY

- CI = estimate ± margin of error = estimate ± (reliability factor) × SE.
- Confidence level $1 - \alpha$: 90%, 95%, 99% correspond to $z^* = 1.645, 1.960, 2.576$.
- The 95% refers to the **method**: about 95% of such intervals capture the true parameter.
- σ known: z interval. σ unknown: t interval with $n - 1$ df. Proportion: $\hat{p} \pm z^* \sqrt{\hat{p}(1 - \hat{p})/n}$.
- Higher confidence or smaller n or larger variability gives a wider interval.
- Plan n from $n = (z^*\sigma/E)^2$; round up.

Practice questions

Try each question before reading the answer underneath it.

Q1. A sample of n = 49 has $\bar{x} = 120$ and $\sigma = 14$ is known. Give the 95% CI.

Answer. SE = 14/7 = 2; margin = 1.96 × 2 = 3.92; CI = [116.08, 123.92].

Q2. A sample of n = 9 has $\bar{x} = 50$, $s = 6$ (σ unknown). Give the 95% CI.

Answer. SE = 2, $t^*(8) = 2.306$, margin = 4.61, CI = [45.39, 54.61].

Q3. Does a 99% CI for the same data come out narrower or wider than the 95% CI?

Answer. Wider, since $z^* = 2.576 > 1.960$.

Q4. Of 400 users, 92 clicked an ad. Give a 95% CI for the click rate.

Answer. $\hat{p} = 0.23$, SE = 0.0210, margin = 0.0412, CI = [0.1888, 0.2712].

Q5. What sample size gives a margin of error of 2 for a mean with $\sigma = 10$ at 95% confidence?

Answer. $n = (1.96 \times 10/2)^2 = 96.04$, so 97.

Q6. A researcher says 'there is a 95% chance the true mean is between 10 and 12'. What is wrong with this?

Answer. The true mean is fixed, not random. The 95% describes the long-run success rate of the method that produced the interval, not a probability for this one interval.